

# Power Electronics Project

Lab: DC-DC Heavy vehicle converter

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# 1 INTRODUCTION

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## 1.1 Design requirements

In this project, we design and build a heavy vehicle DC-DC converter capable of switching a 24V input signal to a 12V, 5A output signal. The circuit requirements are listed below:

- Typical input voltage:  $V_{in} = 24V$  with regulation capabilities in range  $V_{in,min} = 20V$  and  $V_{in,max} = 28V$ .
- Output voltage:  $V_{out} = 12V$ .
- Maximum output current:  $I_{out} = 5A$ .
- Minimum Efficiency: high efficiency is desired, typically exceeding 90% to meet energy-saving goals in heavy vehicle applications.

## 2 DESIGN

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### 2.1 Topology

Several topologies are possible for DC-DC converters, each with specific characteristics and applications. A buck converter (step-down) reduces input voltage to a lower output voltage and is commonly used in high-efficiency, step-down conversion applications. A buck-boost converter provides an output voltage either higher or lower than the input, making it suitable for variable input-output scenarios. The flyback converter employs a transformer for electrical isolation and can step up or step down the voltage, often used in isolated, low-power applications.

For this design, we chose the buck converter due to its widespread use, simplicity, and efficient implementation possibilities. It meets the step-down voltage requirement, as the application involves converting a 24V input to a 12V output, making it an ideal choice. Furthermore, buck converters offer high efficiency for step-down applications, with the possibility to outperform other topologies like buck-boost or flyback converters.

### 2.2 DC-DC converter controller/converter IC

#### 2.2.1 IC Overview and Highlights

The selected IC is the LM3150 Wide-Vin Synchronous Buck Controller from Texas Instruments. This is a high-efficiency synchronous buck controller optimized for step-down voltage conversion in applications like heavy vehicles. This IC has the following characteristics:

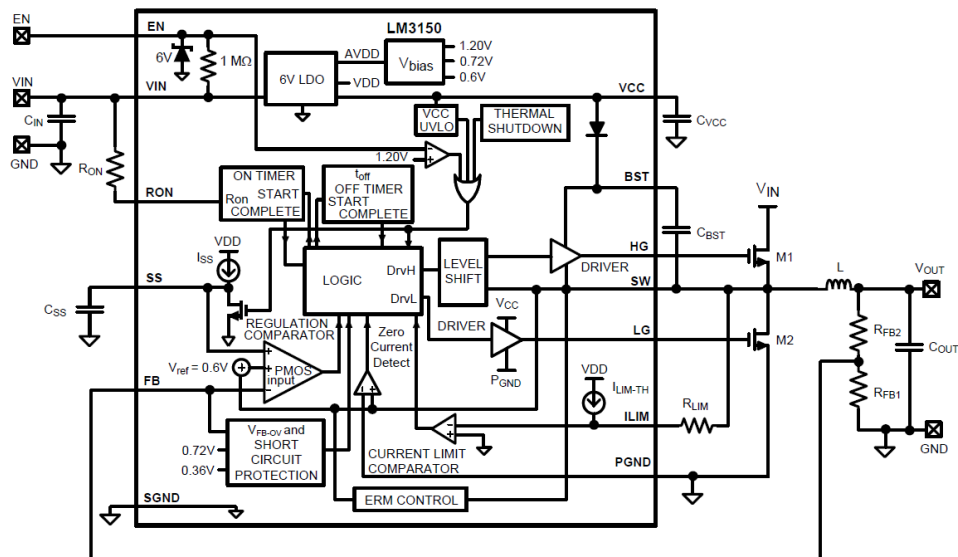
- Wide input voltage range: operates efficiently across a wide input voltage range (6V to 42V), making it suitable for heavy vehicle environments where voltage variations are common.
- High Efficiency: Features synchronous rectification and Constant On-Time (COT) control architecture, enabling efficiencies above 90% under typical load conditions.
- Constant On-Time Control:
  - Ensures fast transient response by adjusting the on-time dynamically based on input voltage and output conditions.
  - Eliminates the need for external loop compensation, simplifying design.
- Emulated Ripple Mode (ERM) Control:
  - Allows the use of low ESR capacitors, reducing cost and output voltage ripple.
  - Achieves precise voltage regulation with minimal output capacitance.
- Integrated Protections:
  - Current Limiting: Detects overload conditions and limits inductor valley current.
  - Short-Circuit Protection: Monitors feedback voltage and reduces output to 0V during a short-circuit condition.
  - Thermal Shutdown: Protects the IC from overheating, ensuring reliability.
- Soft-Start Capability: Gradually ramps the output voltage to its target to avoid inrush currents and minimize startup stress.
- Programmable Switching Frequency:
  - Adjustable switching frequency up to 1 MHz for flexibility between efficiency and component size.
  - Automatically adjusts on-time inversely with input voltage for consistent frequency.

### 2.2.2 Internal Block Diagram

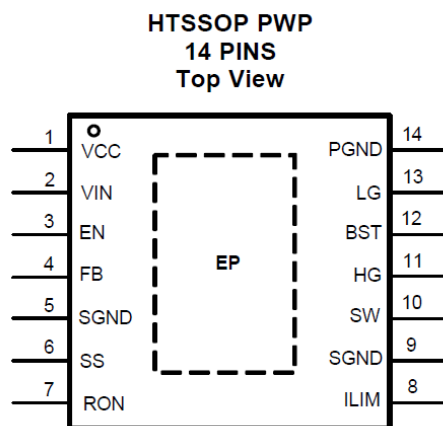
Below we discuss the internal block diagram and functionality. The internal structure of the IC consists of key blocks, including:

- Feedback Control Loop: Compares the feedback voltage (FB) with an internal reference (0.6V) to regulate the output voltage.
- On-Time Generator: Determines the high-side MOSFET's on-time based on the input voltage, output voltage, and external resistor ( $R_{on}$ ).
- Gate Drivers: Drive the external high-side and low-side MOSFETs for synchronous operation, ensuring efficient power delivery.
- Emulated Ripple Mode (ERM) Control: Generates an artificial ripple signal to ensure proper operation with low ESR output capacitors.
- Protection Circuitry: Includes thermal shutdown, short-circuit detection, undervoltage lockout, and current limiting mechanisms.
- Soft-Start Circuit: Controls the rise time of the output voltage by charging an external capacitor ( $C_{SS}$ ) to prevent excessive current draw during startup.

Figure 1 below shows the simplified block diagram of the IC. Figure 2 is a diagram showing the pin configurations and functions.



### Figure 1: Internal Block Diagram of LM3150



### Figure 2: Pin Configuration of LM3150

### 2.2.3 Functionality

The steps below outline the functionality and working principles of the LM3150:

1. **Startup and Initialization:**
  - Upon enabling, the IC checks for undervoltage and overtemperature conditions.
  - The soft-start circuitry gradually ramps the output to the desired voltage.

## 2. Steady-State Operation:

- The feedback pin monitors the output voltage. If it falls below 0.6V, the high-side MOSFET turns on for a fixed duration (on-time).
- After the on-time expires, the low-side MOSFET turns on, and the IC adjusts the switching cycle to maintain regulation.

## 3. Protections:

- If the output current exceeds the defined limit, the controller terminates the on-time early, limiting the inductor current.
- During a short circuit, the IC disables switching and initiates a soft-start cycle after the fault is cleared.

This architecture ensures high efficiency, fast transient response, and reliable operation in demanding heavy vehicle applications.

## 2.3 Component selection

The circuit was designed around the LM3150 IC. For this, we use the LM3150 datasheet to calculate all component values and cross-reference our calculations with the values predicted by Webench to validate our design. The design procedure involves the calculation of the 14 peripheral electronics used around the LM3150 IC as can be seen in the schematic of the heavy vehicle converter in Figure 3.

The first step is to define the power supply operating conditions. The desired regulated output voltage is  $V_{out} = 12V$ . The typical input voltage is  $V_{in,typ} = 24V$ . However, we design the circuit to be able to handle a wider input range with  $V_{in,min} = 20V$  and  $V_{in,max} = 28V$ . We expect the circuit to be able to regulate properly outside this range, but performance will likely begin to suffer. The circuit will obviously not be able to regulate voltages below 12V or above 30V as certain components have a maximum rating of 30V. The circuit is designed to easily handle a load current of 5A,  $I_{load} = 5A$ . Finally, the soft-start time is set to  $t_{ss} = 5ms$ . The soft-start (SS) feature allows the regulator to gradually reach a steady-state operating point, which reduces start-up stresses and current surges.

The second step involves setting the output voltage with feedback resistors. The output voltage is defined as

$$V_{out} = V_{fb} \cdot \frac{R_{fb1} + R_{fb2}}{R_{fb1}}$$

Where  $V_{fb} = 0.6V$ . We set  $R_{fb1}$  to a convenient value ( $R_{fb1} = 10k\Omega$ ) from which we calculate  $R_{fb2} = 191k\Omega$  using the above equation. This feedback network should result in a regulated output voltage,  $V_{out} = 12.06V$ .

The third step is to calculate  $R_{on}$  and the switching frequency,  $f_s$ . The on resistance should be set such that it will allow for the circuit to produce the regulated output voltage with the minimum desired input voltage—in our case, 20V. Here:

$$D_{min} = \frac{V_{out}}{V_{in,min}} = \frac{12}{28} = 0.429$$

$$D_{max} = \frac{V_{out}}{V_{in,max}} = \frac{12}{20} = 0.6$$

$$f_{s,max} = \frac{0.429}{200ns} = 2.14MHz$$

$$t_{off} = \frac{1 - 0.6}{2.14MHz} = 187ns$$

$t_{off}$  should meet the following requirements:

$$t_{off} > t_{off,min} + 200ns$$

$$t_{off} > 387ns$$

At the maximum switching frequency,  $f_s = 2.14MHz$ , the off time is less than  $387ns$  so we need to reduce  $f_s$  so it meets the following condition:

$$f_s < \frac{1 - D}{387ns}$$

$$f_s < 1.03MHz$$

Therefore, a switching frequency is arbitrarily chosen at  $500kHz$  which should allow for reasonable size components and satisfies the requirements above. Therefore,  $f_s = 500kHz$ . Using this,  $R_{on}$  can be calculated:

$$R_{on} = \frac{(V_{out} \cdot V_{in}) - V_{out}}{V_{in} \cdot K \cdot f_s} + R_{ond}$$

Where:

$$R_{ond} = -((V_{in} - 1) \cdot (V_{in} \cdot 16.5 + 100)) - 1000$$

$R_{on}$  is then set to a standard value from the datasheet such that the minimum input voltage of  $20V$  still results in the desired output voltage. From this, we select  $R_{on} = 205k\Omega$ .

The fourth step involves calculating the inductor required.

$$ET = (24 - 12) \cdot (12/24) \cdot (1000/500) = 12V\mu s$$

From the inductor nomograph in the datasheet, a  $5A$  load and  $12V\mu s$  calculation corresponds to a  $L06$  type of inductor which is a  $6.8\mu H$  inductor. To ensure the inductor is adequately rated, we add a 20% safety margin therefore opting for an inductor,  $L_1 = 8.2\mu H$ .

Once the inductor has been selected, we can proceed to select the output capacitance,  $C_{out}$ . The voltage rating on the output capacitor should be greater than or equal to the output voltage. As a rule of thumb most capacitor manufacturers suggest not to exceed 90% of the capacitor rated voltage. In the case of multilayer ceramics, the capacitance will tend to decrease dramatically as the applied voltage is increased towards the capacitor rated voltage. The capacitance can decrease as much as 50% when the applied voltage is only 30% of the rated voltage. The chosen capacitor should also be able to handle the *rms* current which is equal to:

$$I_{rmsco} = I_{out} \cdot \frac{r}{\sqrt{12}}$$

For this design, the chosen ripple current ratio,  $r = 0.3$ , represents the ratio of inductor peak-to-peak current to load current  $I_{out}$ . A good starting point for ripple ratio is 0.3 but it is acceptable to choose  $r$  between 0.25 to 0.5. The nomographs in the datasheet all use 0.3 as

the ripple current ratio. Therefore, the capacitor should be able to handle  $I_{rmsco} = 0.433A$ . Additionally, the maximum ESR allowed to prevent overvoltage protection during normal operation is  $54m\Omega$ . With this set of restrictions,  $C_{out} = 56\mu F$  and is a polymer aluminium capacitor.

From Step 5, the capacitor chosen in ESR is small enough that we should use a feed-forward capacitor. This is calculated by:

$$C_{ff} = \frac{V_{out}}{V_{in,min} \cdot f_s} \cdot \frac{R_{fb1} + R_{fb2}}{R_{fb1} \cdot R_{fb2}} = 126pF$$

We let  $C_{ff} = 130pF$ , which is the next standard value.

The next step is MOSFET and  $R_{lim}$  selection. The LM3150 controller is designed to drive N-channel MOSFETs. For a maximum input voltage of  $28V$  we should choose N-channel MOSFETs with a maximum drain-source voltage,  $V_{ds}$ , greater than  $(1.2 \cdot 28 = 33.6V)$ . FETs with maximum  $V_{ds}$  of  $40V$  will be the first option. The combined total gate charge  $Q_{g,total}$  of the high-side and low-side FET should satisfy the following:

$$Q_{g,total} \leq I_{vccl}/f_s$$

$$Q_{g,total} \leq 65mA/500kHz$$

$$Q_{g,total} \leq 130nC$$

Along with considering power consumption requirements, we opt for the high side MOSFET,  $M1$ , being the CSD18504Q5A with  $V_{ds,max} = 40V$  and  $I_{ds,max} = 50A$ . The low side MOSFET,  $M2$ , is chosen to be CSD18537NQ5A with  $V_{ds,max} = 60V$  and  $I_{ds,max} = 50A$ .

The next step is to calculate the input capacitance. The input capacitor should be chosen so that the voltage rating is greater than the maximum input voltage which for this circuit is  $28V$ . Similar to the output capacitor, the voltage rating needed will depend on the type of capacitor chosen. The input capacitor should also be able to handle the input  $rms$  current, which is a maximum of approximately  $0.5 \cdot I_{out} = 2.5A$ . The minimum capacitance with a maximum 5% input ripple  $\Delta V_{in,max} = (0.05 \cdot 28) = 1.4V$  is:

$$C_{in} = (12 \cdot 0.275 \cdot (1 - 0.275)) / (500kHz \cdot 0.6) = 15.95\mu F$$

To handle the large input rms current, 3 ceramic capacitors are chosen at  $4.7\mu F$  each with a voltage rating of  $50V$ . Each ceramic capacitor is capable of handling  $3A$  of  $rms$  current. Another capacitor of 5 times the combined input capacitance ( $5 \cdot 16\mu F = 80\mu F \approx 100\mu F$ ) is chosen to provide input voltage filter damping because of the low ESR ceramic input capacitors. The bypass capacitor,  $C_{byp} = 0.1\mu F$ , is a ceramic with a voltage rating greater than maximum  $V_{in}$ .

The ninth step in the design process is to calculate the soft-start capacitor. The soft start-time should be greater than the input voltage rise time and satisfy the equality below to maintain a smooth transition of the output voltage to the programmed regulation voltage during startup. The desired soft-start time,  $t_{ss}$ , of  $5ms$  must satisfy:

$$(t_{ss} \geq (V_{out} \cdot C_{out}) / (I_{ocl} - I_{out}))$$

by using the chosen component values through the previous steps, this leaves us with a soft-start capacitor equal to  $C_{ss} = 15nF$ . Here, we choose a ceramic capacitor.

To cross-validate our calculations based on the datasheet, we use the TI Webench tool to support the design of the heavy vehicle converter. Webench applies some rounding to electronic component values to the nearest standard values and also does some fine-tuning to the switching frequency. Overall however, our calculation broadly agree with those set by Webench. The schematic in Figure 3 below shows the final DC-DC converter.

## 2.5 Bill of Materials

### Table 1: LM3150 24V-12V DC-DC Converter Bill of Materials

| Part | Manufacturer | Part Number         | Farnell Code | Quantity Required | Order Quantity |
|------|--------------|---------------------|--------------|-------------------|----------------|
| Cff  | MURATA       | GRM21A5C2E470JW01D  | 2218866      | 3                 | 10             |
| Cbst | Panasonic    | ECPU1C474MA5        | 9694293      | 1                 | 5              |
| Cin  | MURATA       | GRM31CR71H475KA12L  | 1735545      | 3                 | 5              |
| Css  | TDK          | C2012C0G1H153J085AA | 1907326      | 1                 | 10             |
| L1   | Vishay-Dale  | IHLP2525CZER8R2M01  | 3518689      | 1                 | 5              |
| M1   | TI           | SI3456DDV-T1-GE3    | 1840419      | 1                 | 5              |
| U1   | TI           | LM3150MH/NOPB       | 3004219      | 1                 | 1              |
| Cout | Panasonic    | EEHZC1E560P         | 2095328      | 1                 | 1              |
| Cbyp | Kemet        | C0805C104M5RACTU    | 1650866      | 1                 | 10             |
| Cvcc | MURATA       | GRM21BR71A225KA01L  | 4327075      | 1                 | 5              |
| M2   | TI           | SI3460DDV-T1-GE3    | 1858948      | 1                 | 5              |

### 3 SIMULATION

In this section, we provide some simulation results provided by TI. The figures below show startup, load transient, input transient, and steady state respectively. The main takeaways from the simulation results is the soft-start (1ms in simulation versus 5ms in practice). An RMS output voltage of 12.6V (discussed in section 5). Simulation results predict a steady-state efficiency of approximately 95% at a switching frequency of 514.55kHz. Finally,  $V_{out}$  has a no load tolerance of 3.96% with a peak switch current,  $I_{sw,peak} = 5.84A$ .

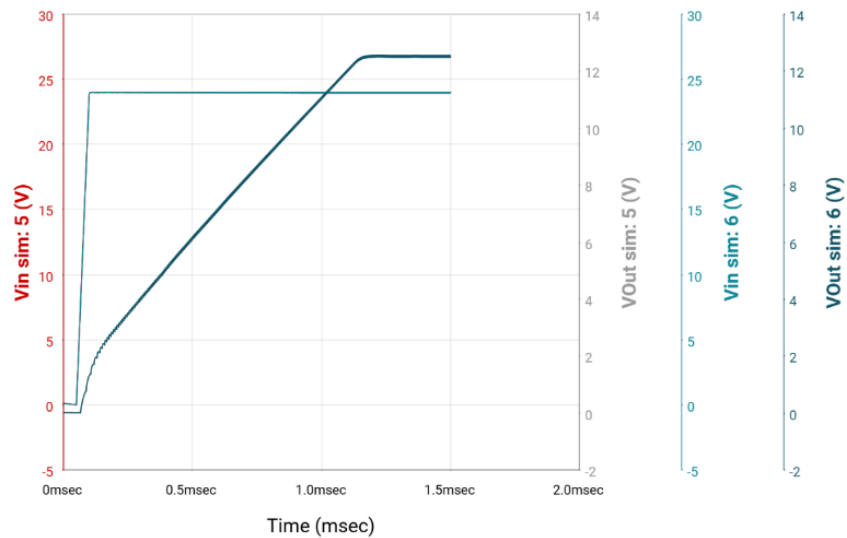


Figure 4: LM3150 24V-12V DC-DC Converter Startup

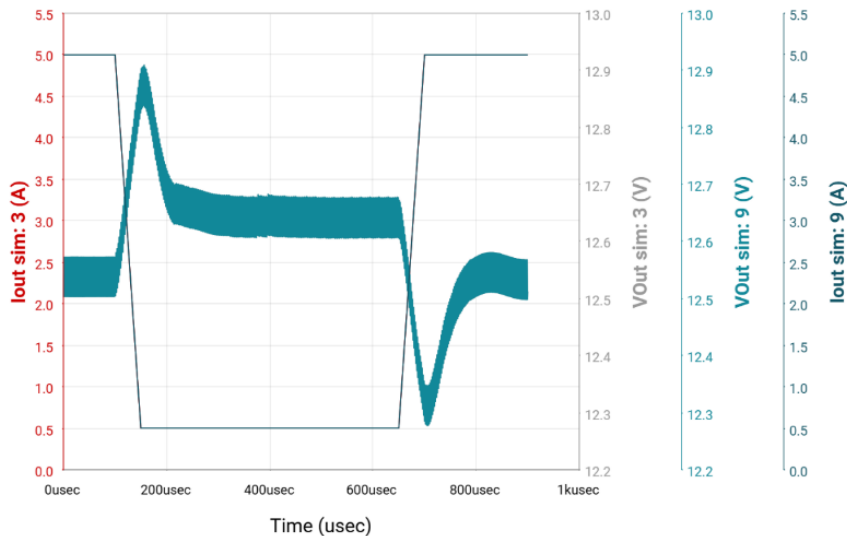
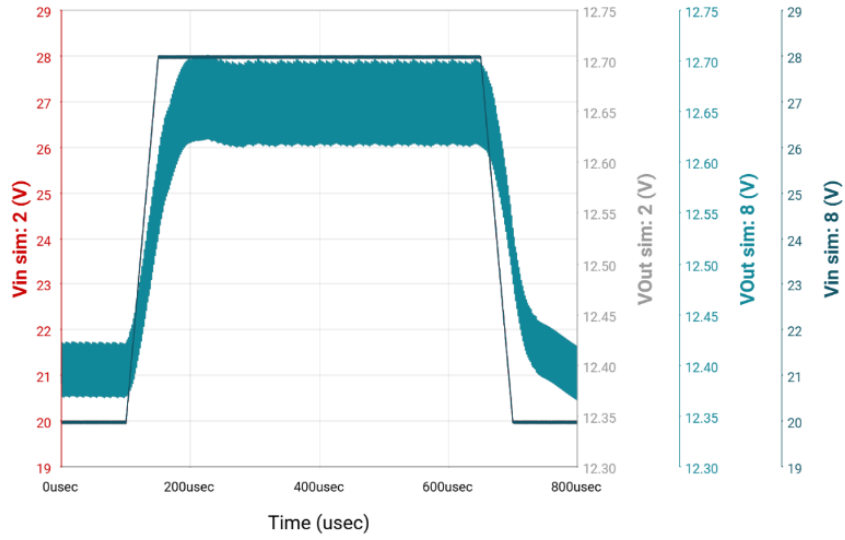
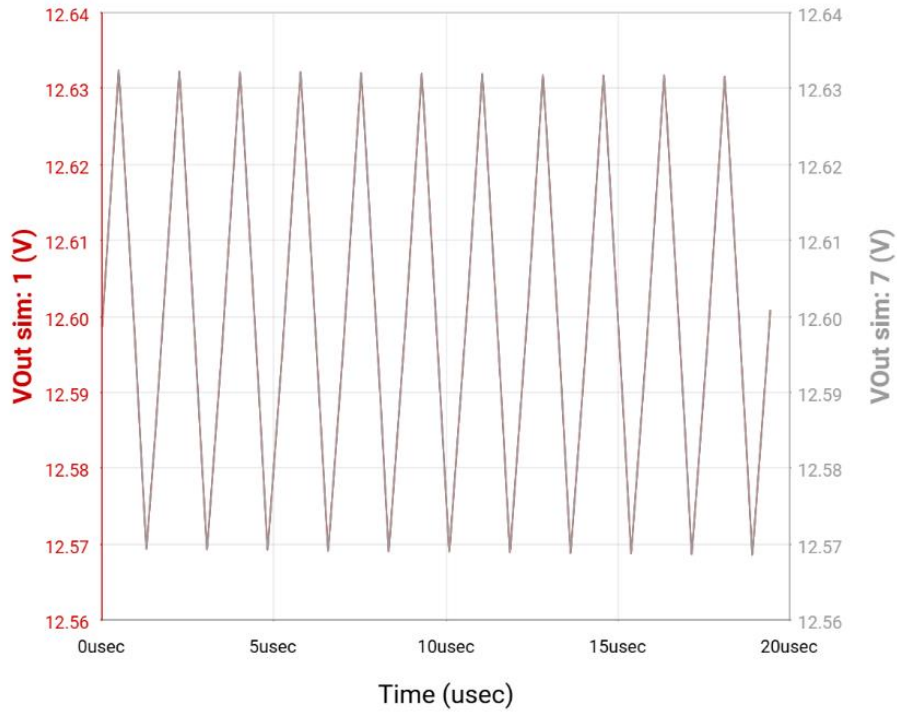


Figure 5: LM3150 24V-12V DC-DC Converter Load Transient



**Figure 6: LM3150 24V-12V DC-DC Converter Input Transient**



**Figure 7: LM3150 24V-12V DC-DC Converter Steady State**

### 3.1 PCB

The figures below show the top and bottom view of our PCB. In general, the objective is to minimize parasitic inductance and prevent high  $dv/dt$  or high  $di/dt$  currents from contaminating sensitive signal. In this design we therefore keep the gate drive traces short, i.e. the traces from  $HG$  and  $SG$  to the gate of the high-side and low-side MOSFETs respectively. The shorter traces help in avoiding picking up noise and ensure switching is fast. In this circuit, the high-side gate driver voltage is supplied by the differential voltage between the  $BST$  pin and  $SW$  pin. We connect a capacitor  $C_{bst}$  which creates the voltage used to drive the gate of

the high-side FET. We keep this capacitor physically close to the corresponding pin to keep the current paths short and to minimize the trace inductance. A similar reasoning is applied for the input decoupling capacitors. Finally, it is critical that the switch node be made only as large as required to handle the load current. There are fast voltage transitions occurring in the switch node at a high frequency, and if the switch node is made too large it may act as an antennae and couple switching noise into other parts of the circuit. We therefore keep this node as small as possible and make any connections to the switch pin as small as possible.

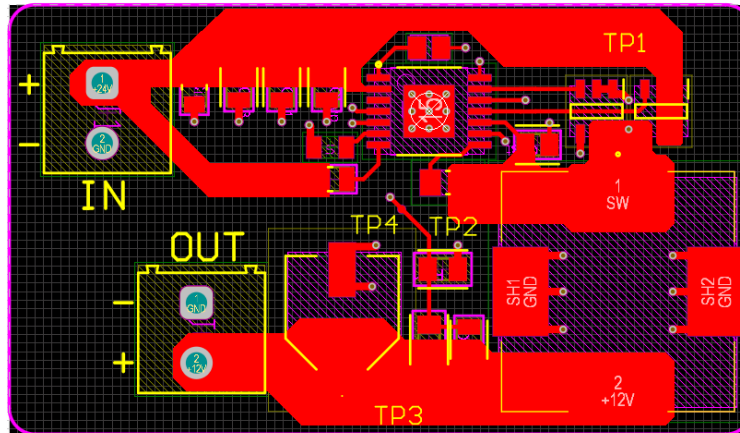


Figure 8: LM3150 24V-12V DC-DC Converter PCB Top Layer

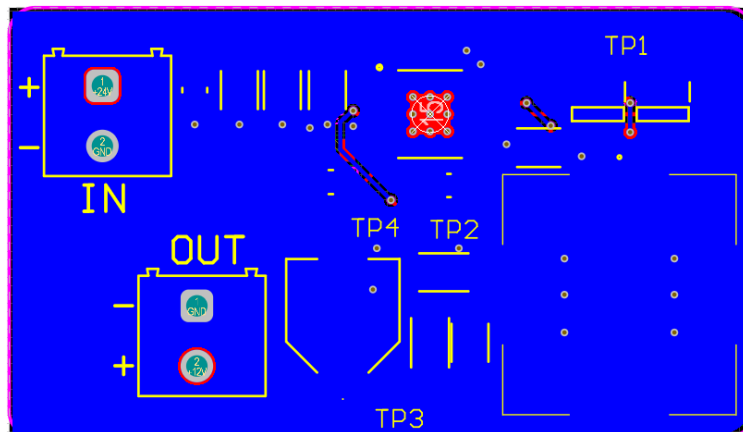


Figure 9: LM3150 24V-12V DC-DC Converter PCB Bottom Layer

## 4 MATERIALS

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### 4.1 Converter

The image below shows the final constructed PCB. The LM3150 can be seen in the top center region of the PCB with the high-side and low-side MOSFETs directly to the right of it. Four test pins have been placed across the PCB to measure input voltage, switching voltage, feedback voltage, and output voltage.

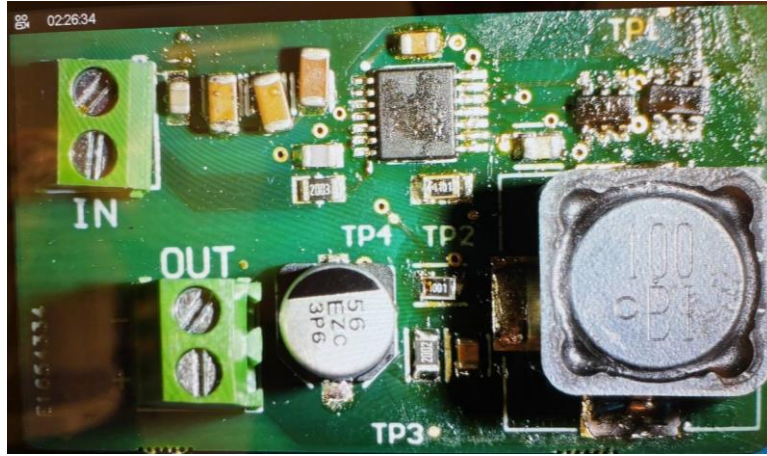


Figure 10: LM3150 24V-12V DC-DC Converter PCB

## 4.2 Measuring Instruments

We use the following measuring instruments in the laboratory:

- Oscilloscope
- Power Supply
- Electronic Load
- Multimeter
- Thermal Camera
- Soldering Station
- LCR Meter
- Current Probe

## 5 RESULTS

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The actual implemented circuit on the PCB uses slightly different components values to the ones calculated and shown in Section 2. Firstly, we decreased the value of both feedback resistors by one order of magnitude. This should have no effect on the regulated output voltage but will result in more current flowing through the feedback network. Additionally, we use a slightly higher resistance value for  $R_{fb2}$  ( $20k\Omega$  instead of  $19.1k\Omega$ ). We therefore expect to get a slightly higher regulated output voltage (around  $12.6V$ ) which can be seen in the subsequent measurements and simulation. We also use a slightly smaller feed-forward capacitor ( $C_{ff} = 100pF$ ). This has no major implications for the final circuit. Additionally, we use a slightly higher rated inductor. This also has no major implications for the final circuit.

More specifically regarding the feedback network, the regulated output voltage is set by the two external feedback resistors  $R_{fb1}, R_{fb2}$  by:

$$V_{out} = V_{fb} \cdot \frac{R_{fb1} + R_{fb2}}{R_{fb1}}$$

Where the feedback voltage,  $V_{fb}$ , at  $FB$  is compared to the internal reference voltage of  $0.6V$ . In the original design, with  $R_{fb1} = 10k\Omega$  and  $R_{fb2} = 191k\Omega$ , we expect the output voltage to be regulated to  $V_{out} = 12.06V$ . With the alternative feedback resistors set to  $R_{fb1} = 1k\Omega$  and  $R_{fb2} = 20k\Omega$ , we expect the output voltage to be regulated to  $V_{out} = 12.60V$ .

Finally, in our design, we decided to go for less “overkill” MOSFETs and decided to go for the SI3460DDV-T1-GE3 MOSFET as both our low-side and high-side MOSFET. This selection should have no significant impact on the output voltage.

## 5.1 Static load regulation

The figure and table below show the output voltage of the circuit at different loads for an input voltage of  $24V$ . As we can see, the regulated output voltage remains stable and close to  $12.7V$  as expected given the change in feedback resistance. Additionally, even when operating with a large load, the current does not exceed that maximum  $5A$  rating.

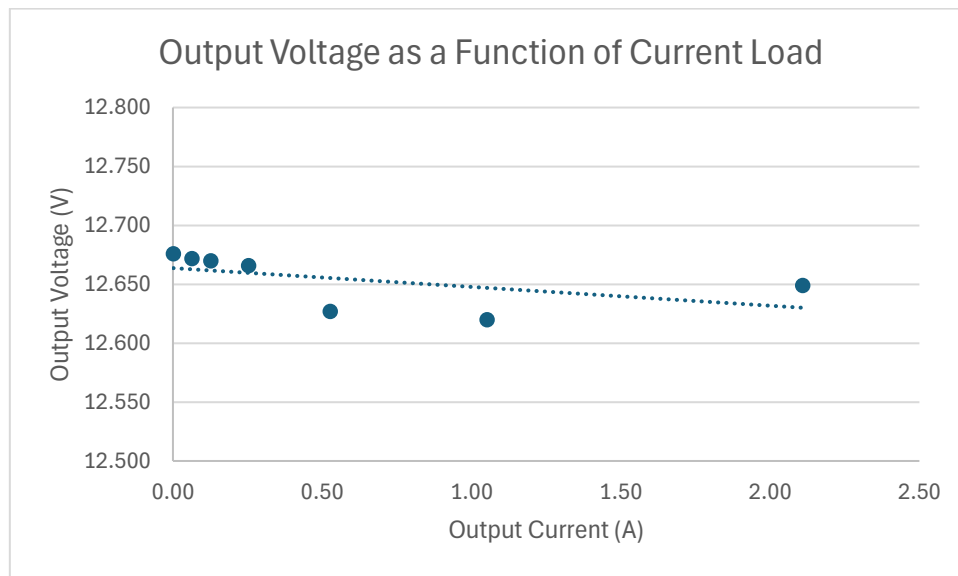


Figure 11: LM3150 24V-12V DC-DC Converter Static Load Regulation

Table 2: LM3150 24V-12V DC-DC Converter Static Load Regulation Measurements

| $Load(\Omega)$ | $V_{in} (V)$ | $V_{out} (V)$ | $I_{in} (A)$ | $I_{out}(A)$ |
|----------------|--------------|---------------|--------------|--------------|
| No Load        | 24.00        | 12.672        | 0.025        | —            |
| 6              | 24.00        | 12.649        | 0.310        | 2.1082       |
| 12             | 24.00        | 12.620        | 0.587        | 1.0517       |
| 24             | 24.00        | 12.627        | 0.307        | 0.5261       |
| 50             | 24.00        | 12.666        | 0.162        | 0.2533       |
| 100            | 24.00        | 12.670        | 0.093        | 0.1267       |
| 200            | 24.00        | 12.672        | 0.059        | 0.0634       |
| 10000          | 24.00        | 12.676        | 0.032        | 0.0013       |

We evaluate the accuracy and stability of the output voltage across different load resistances while maintaining a constant input voltage of  $24V$ . The output voltage remains highly stable across a wide range of load resistances, showing minimal deviation from the nominal value

The results confirm the effectiveness of the voltage regulation circuit in maintaining consistent output, even under varying load conditions. The DC-DC converter performs as expected, maintaining stable output voltage across all tested loads, ensuring compatibility with a variety of loads.

## 5.2 Input regulation

We now turn our attention to input regulation to see how the circuit handles different input voltages. For this, we test our circuit for an input voltage range of 20V to 28V. The graph and table below show the measurements performed. We carried out measurements a 200Ω and no load setup.

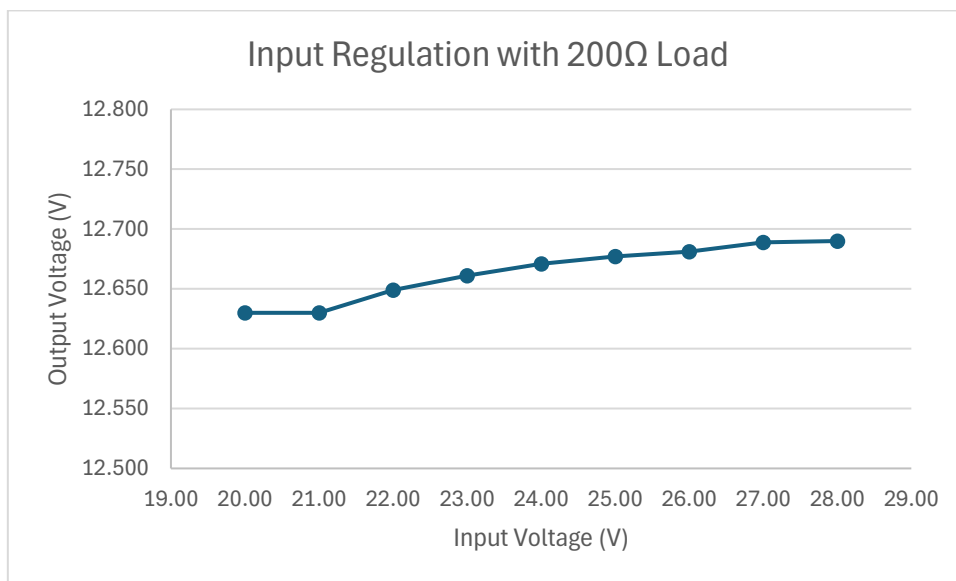
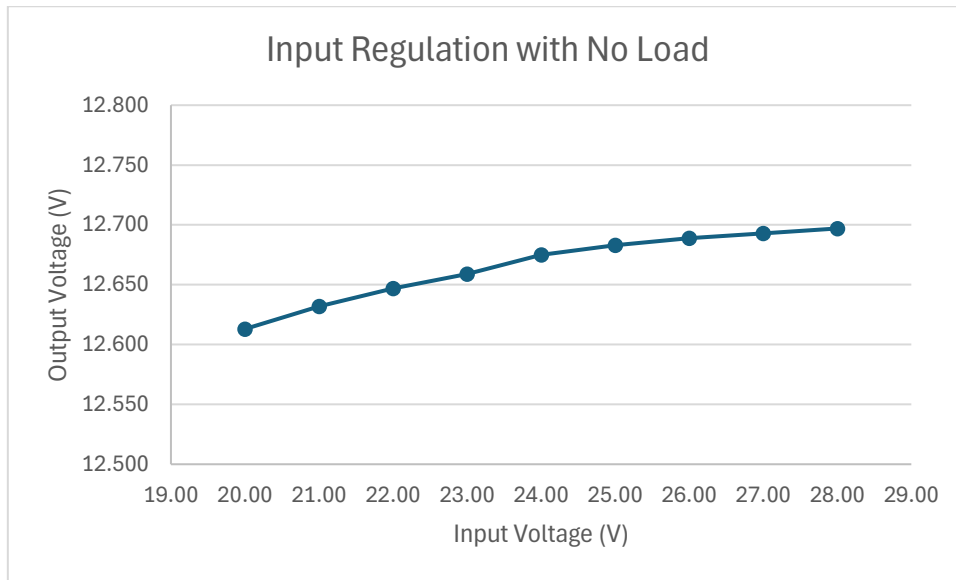


Figure 12: LM3150 24V-12V DC-DC Converter Input Regulation 200Ω

Table 3: LM3150 24V-12V DC-DC Converter Input Regulation Measurements

| $V_{in}(V)$ | $V_{out}(V)$ | $V_{switch}(V)$ |
|-------------|--------------|-----------------|
| 20.00       | 12.630       | 12.630          |
| 21.00       | 12.630       | 12.649          |
| 22.00       | 12.649       | 12.663          |
| 23.00       | 12.661       | 12.677          |
| 24.00       | 12.671       | 12.688          |
| 25.00       | 12.677       | 12.697          |
| 26.00       | 12.681       | 12.702          |
| 27.00       | 12.689       | 12.703          |
| 28.00       | 12.690       | 12.712          |



**Figure 13: LM3150 24V-12V DC-DC Converter Input Regulation No Load**

**Table 4: LM3150 24V-12V DC-DC Converter Input Regulation No Load Measurements**

| $V_{in}(V)$ | $V_{out}(V)$ | $V_{switch}(V)$ |
|-------------|--------------|-----------------|
| 20.00       | 12.613       | 12.629          |
| 21.00       | 12.632       | 12.645          |
| 22.00       | 12.647       | 12.661          |
| 23.00       | 12.659       | 12.673          |
| 24.00       | 12.675       | 12.684          |
| 25.00       | 12.683       | 12.690          |
| 26.00       | 12.689       | 12.702          |
| 27.00       | 12.693       | 12.708          |
| 28.00       | 12.697       | 12.714          |

As we can see, the circuit performs well across the range of input voltages with only a small difference of  $0.060V$  and  $0.084V$  between  $V_{in} = 20V$  and  $V_{in} = 28V$  for the  $200\Omega$  and no-load setup respectively. A small upward creep in the output as the input goes from  $20V$  up to  $28V$  is fairly normal in practice and is usually due to a combination of factors such as non-ideal line regulation, conduct and switching losses, and feedback network tolerances. All of these add up to the few millivolts increases we see across our measurements. Essentially, the controller is regulating properly but the circuit components are not perfectly ideal meaning a slight rise in  $V_{out}$  at higher  $V_{in}$  is expected and within normal line regulation behavior.

### 5.3 Internal working

In this section we take a deeper look at the switching behavior of the IC  $200\Omega$  setup. Figure 14 below shows the signal at the switching pin ( $CH1$ ) and

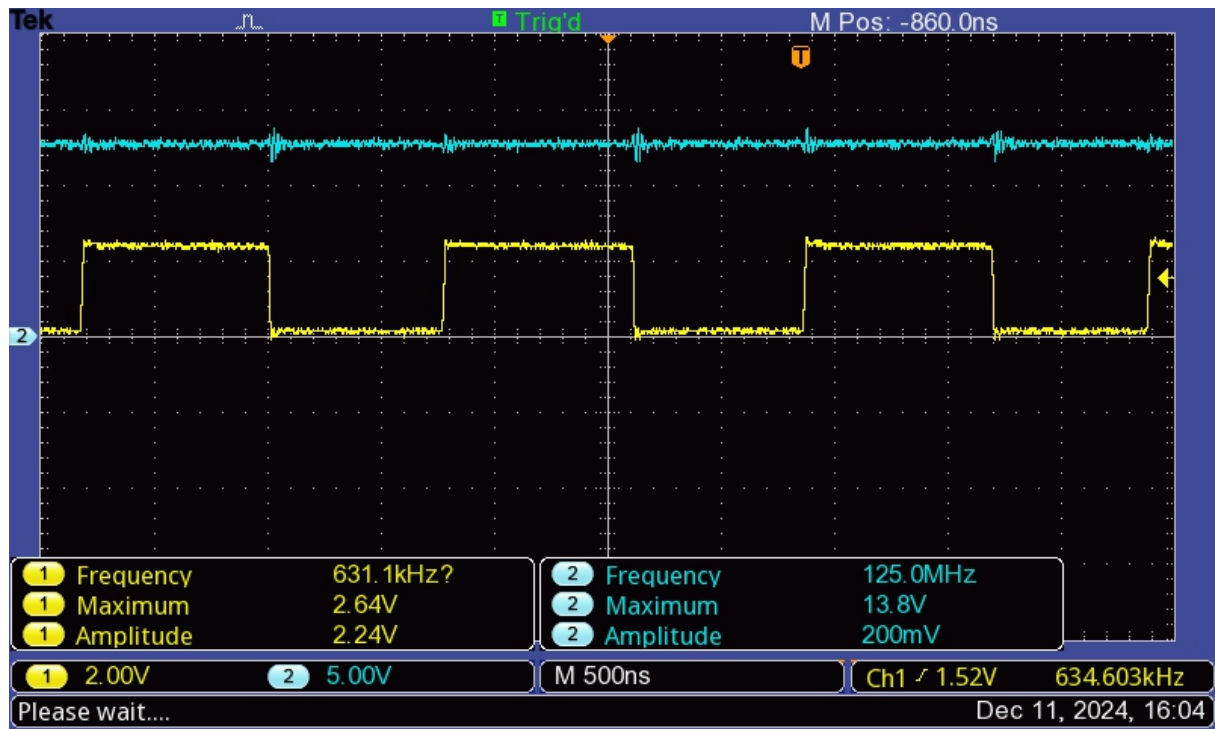


Figure 14: LM3150 24V-12V DC-DC Converter 200Ω Switching Signal

Table 5: LM3150 24V-12V DC-DC Converter 200Ω Measurements

| $V_{in}(V)$ | $V_{out}(V)$ | $V_{switch}(V)$ |
|-------------|--------------|-----------------|
| 20.00       | 12.630       | 12.630          |
| 21.00       | 12.630       | 12.649          |
| 22.00       | 12.649       | 12.663          |
| 23.00       | 12.661       | 12.677          |
| 24.00       | 12.671       | 12.688          |
| 25.00       | 12.677       | 12.697          |
| 26.00       | 12.681       | 12.702          |
| 27.00       | 12.689       | 12.703          |
| 28.00       | 12.690       | 12.712          |

The switch pin measurements confirm that the converter operates reliably. The measurements also demonstrate reliable regulation, maintaining a nearly constant output voltage across the tested input voltage range.

For  $V_{in} = 24V$ ,  $V_{out} = 12.671$  and using the formulas from the datasheet:

$$D = t_{on} \cdot f_s = 850 \cdot 10^{-9} \cdot 631 \cdot 10^3 = 53.6\%$$

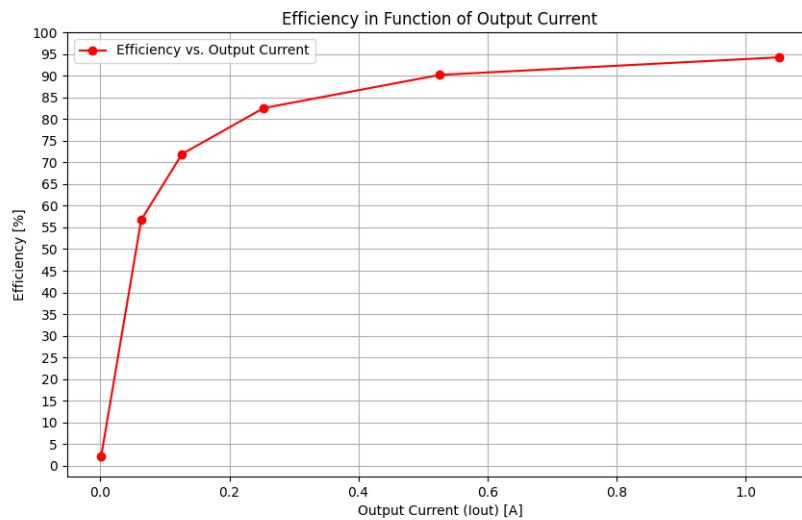
Or alternatively:

$$D \approx \frac{V_{out}}{V_{in}} = \frac{12.671}{24} = 52.8\%$$

The IC is operating in Continuous Conduction Mode (CCM). The inductor current does not reach zero during the OFF phase, as the load ( $200\Omega$ ) is moderate. This suggests operation in CCM.

## 5.4 Efficiency

Efficiency is an important consideration when designing electronic circuits and particularly when designing DC-DC converter circuits such as the one in this project. The figure below plots the efficiency of the buck converter as a function of output current. The theoretical efficiency of the converter during simulation is around  $\eta = 95\%$ , which we see we get close to in practice ( $\approx 94.2\%$ ). The efficiency curve plotted below follows the behavior we expect at low currents, a larger proportion of the input power is consumed by the converter's internal losses rather than being delivered to the load, resulting in lower efficiency. As the output current increases, these overhead losses become a smaller fraction of the total power, causing the efficiency to climb before plateauing near its maximum value. Ultimately, the measured performance aligns closely with the expected behavior, demonstrating that the design meets its efficiency targets.



**Figure 15: LM3150 24V-12V DC-DC Efficiency**

Efficiency is calculated by using the measurements for input and output voltage and current:

$$\eta = \left( \frac{V_{out} \cdot I_{out}}{V_{in} \cdot I_{in}} \right) \cdot 100$$

## 5.5 Ripple and noise

It is important to be aware of AC ripple and noise in the signal as in an ideal scenario these phenomena would not be present. Figures 16 – 18 below give an idea of these signals and how they influence the DC-DC converter.

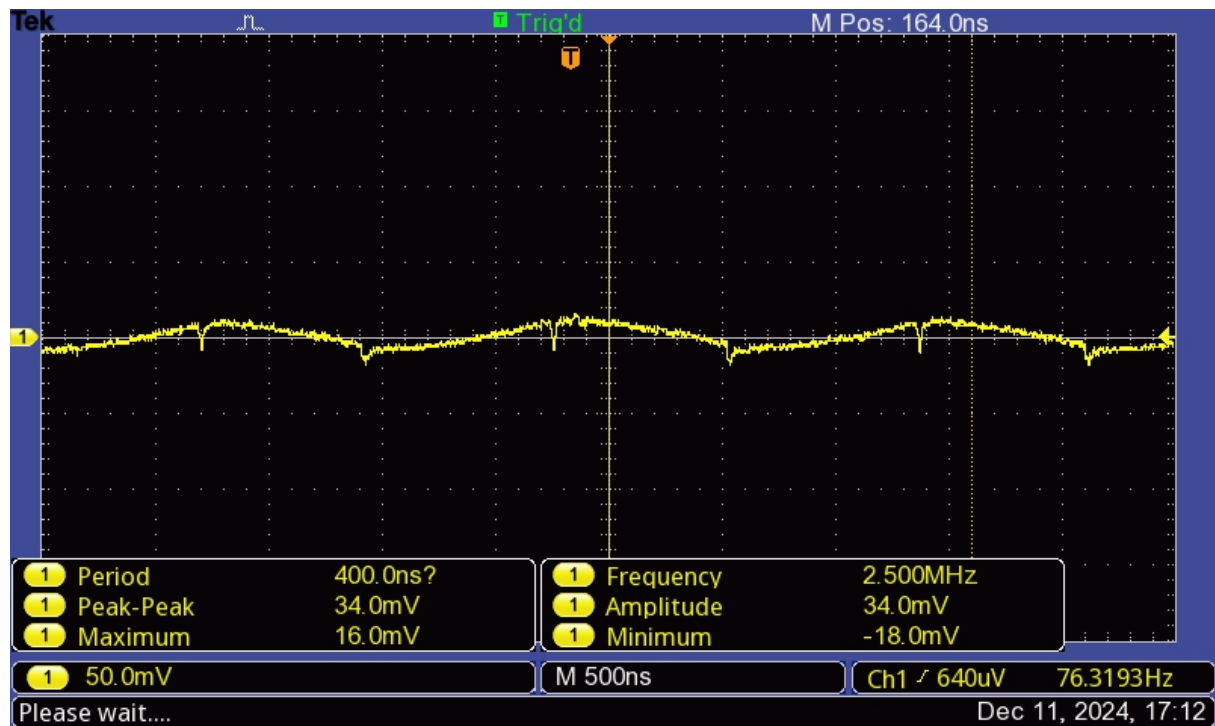


Figure 16: LM3150 24V-12V DC-DC Converter Ripple

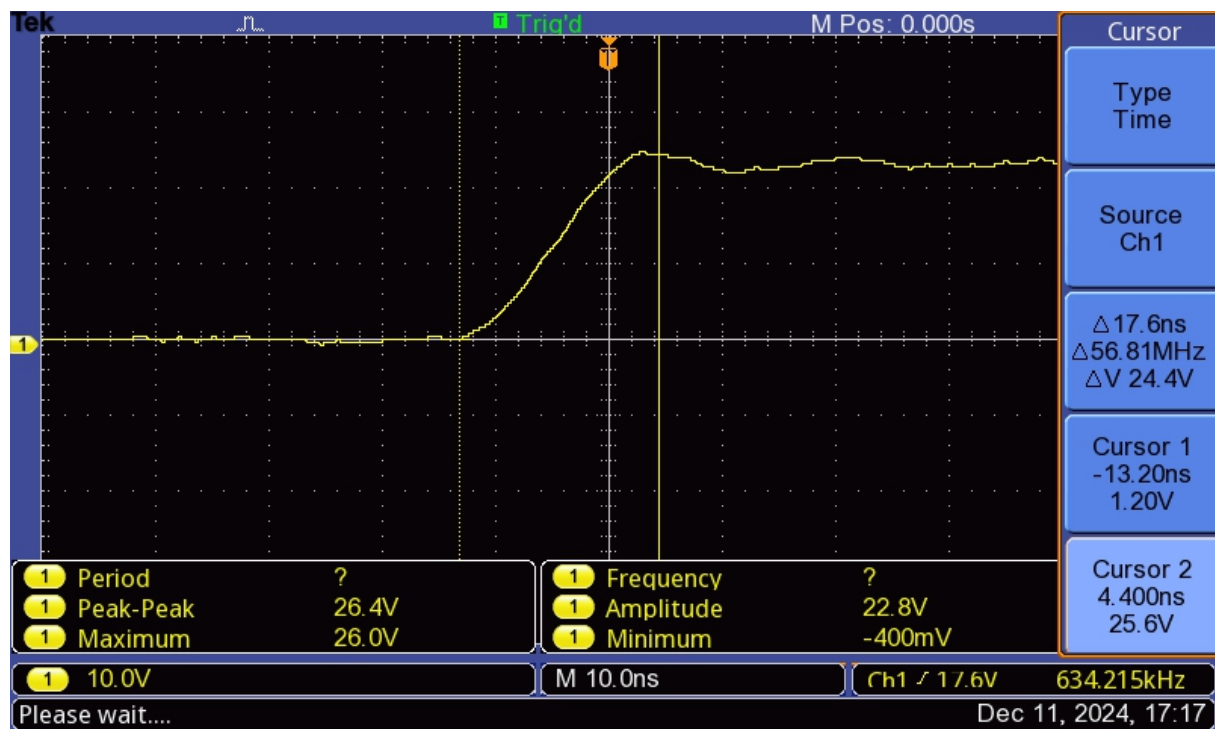


Figure 17: LM3150 24V-12V DC-DC Converter Switching

The ripple observed in the first oscilloscope screenshot shows Peak-to-Peak Ripple voltage of 34 mV and Frequency of 2.5 MHz. This is within acceptable limits. The ripple waveform

indicates the presence of switching noise with a relatively low amplitude, suggesting that the filtering components are functioning effectively.

From the switching waveform in Figure 17, the rise time of the switching node is approximately  $17.6\text{ ns}$ . The voltage swing is  $26.4/10 = 2.64\text{V}_{pp}$  (10x attenuation) and the switching frequency is  $f_s = 634.215\text{ kHz}$ .

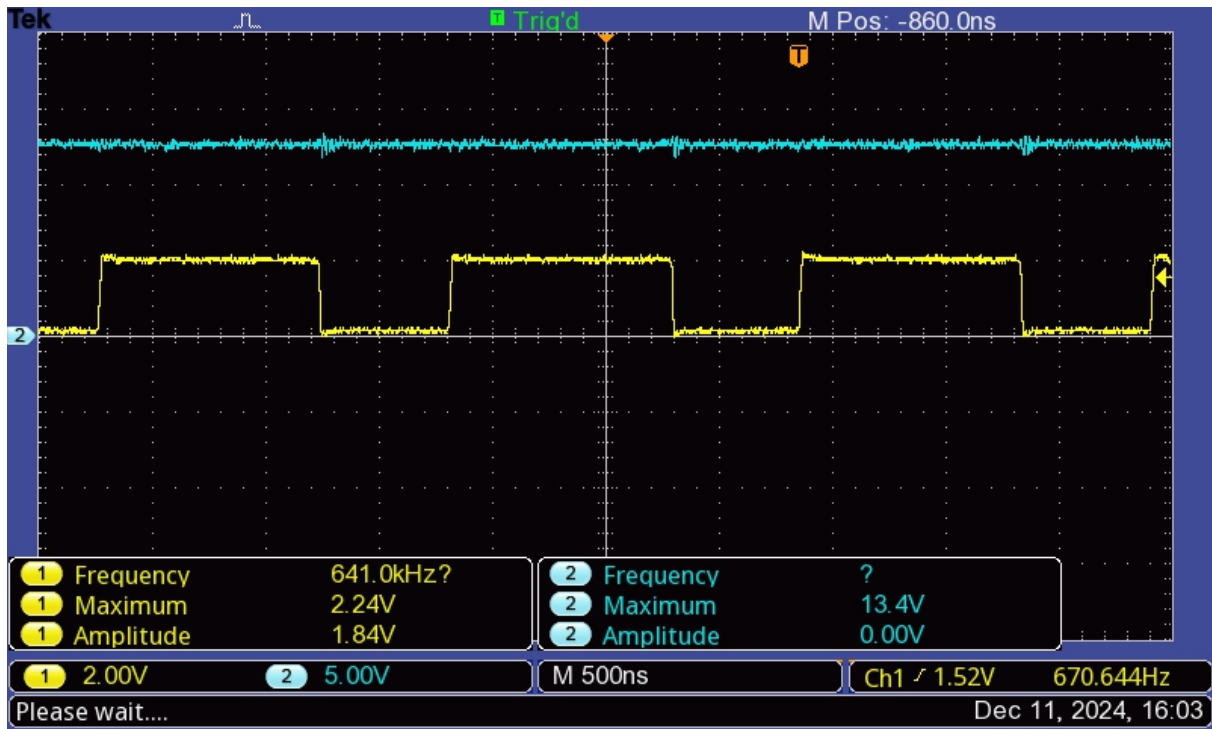


Figure 18: LM3150 24V-12V DC-DC Converter 200Ω Switching Signal

## 5.6 Dynamic load

Beyond static testing, it is equally important to test the circuit in a dynamic load configuration as in real-life applications, many loads are dynamic, varying in response to operating conditions. When a load changes suddenly, power supplies generally exhibit overshoot when the load decreases and undershoot when it increases. Key parameters, such as settling time and peak voltage deviations, provide insight into the supply's ability to respond effectively to these abrupt changes, highlighting its stability and responsiveness under real-world conditions. To perform a dynamic load test, we build the circuit shown below where the 12V regulated voltage (supply under test) is our heavy vehicle converter. We perform this test with a 10V square wave at  $250\text{ kHz}$ . The figures below show the output signal of the DC-DC converter at this frequency.

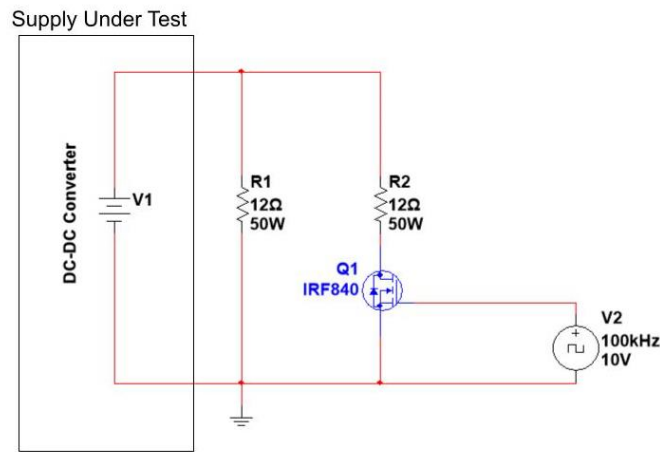


Figure 19: Dynamic Load Test Circuit Setup

We expect the supply to overshoot if the load is suddenly reduced and undershoot if the load is suddenly increased, which we see in the figure below. What is interesting to see however is that when the supply overshoots, there is also ringing on the signal, whereas when the supply undershoots, there is a single peak, i.e. no ringing. The explanation we have for this is that when the load decreases, the inductor still carries a high current, but the load is suddenly lighter, so the excess energy in the LC filter causes the output to shoot above nominal. Resonance among the inductor, output capacitor, and parasitic elements shows up as ringing until the circuit settles. Conversely, when the load increases, the output voltage experiences a dip because there is not immediately enough inductor current to supply the heavier load. The control loop ramps the duty cycle and inductor current, bringing the voltage back up. This typically produces a single undershoot “peak” (rather than extended ringing) as the converter transitions to the higher current state. We note however that the overshoot peak is significantly larger than the undershoot peak signaling that the reason could lie elsewhere in the design beyond the explanation we have provided.

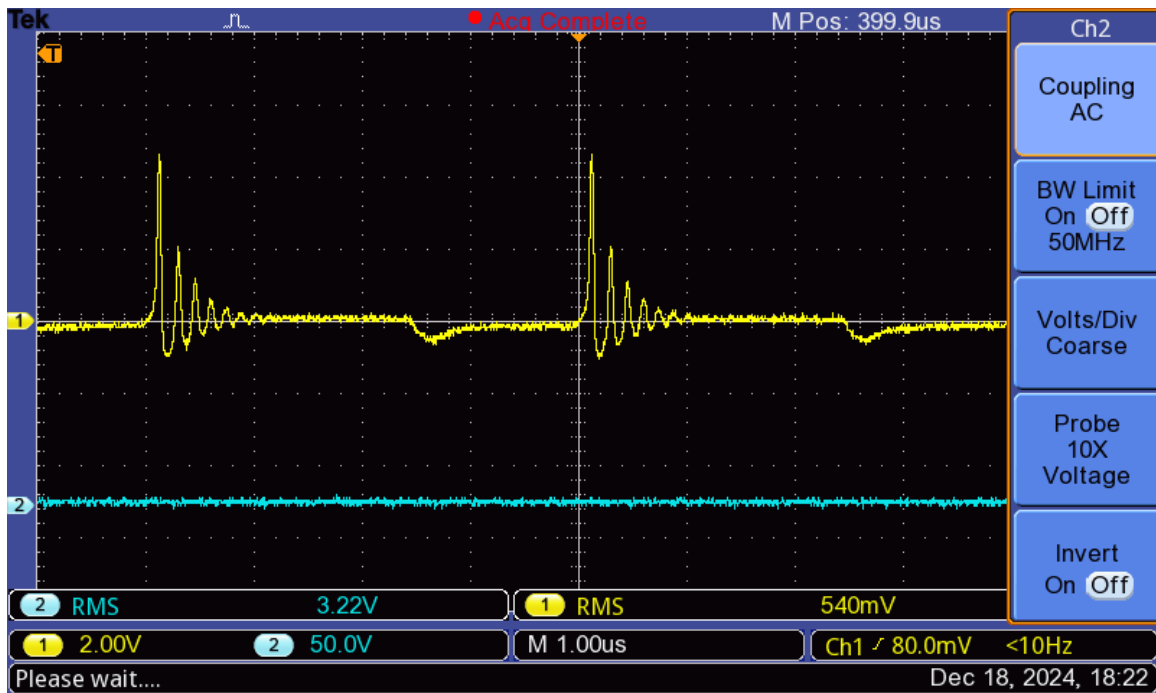


Figure 20: LM3150 24V-12V DC-DC Converter Dynamic Load Test

During overshoot, the circuit settles after approximately  $t_{set,over} = 0.8\mu s$  with an oscillating frequency of  $f_{osc,over} = 6MHz$ . During undershoot, the output voltage dips approximately  $\Delta V_{peak,under} = 0.4V$ .

## 5.7 Temperature

Using an IR camera, the surface temperatures of the IC and MOSFETs were measured after 5 minutes of operation at  $V_{in} = 24V$ ,  $I_{in} = 0.059A$  and using a  $200\Omega$  load. The LM3150 IC reached a temperature of  $44.1^{\circ}C$ . This indicates efficient thermal management, as the temperature is well below the thermal shutdown threshold of  $165^{\circ}C$ . The switching MOSFETs reached a temperature of  $40.9^{\circ}C$ .

## 6 DISCUSSION / CONCLUSION

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The purpose of this lab project was to design and test a DC-DC converter that we can use for heavy vehicle applications. Our main objectives were:

1. Maintain a stable regulated output voltage of  $12V$  for a  $24V$  input voltage.
2. Achieve high efficiency, ideally exceeding 90% to meet the energy saving requirements.
3. To ensure that the converter can operate under different load conditions from light to heavy loads.

After testing and measuring the following are key results and insights:

1. Voltage regulation: Across different loads, from no-load to high load, the output voltage remained consistently around  $12.6V$ , showing minimal deviation. The deviations noted are within expectations and reasonable functioning of such circuits. Additionally, during input regulation, the output remained stable and within design expectations.
2. Switching behaviour: at moderate and high loads the inductor remained continuous indicating the IC was operating in continuous conduction mode. The duty cycle closely matched theoretical calculations which shows that the controller was regulating the output as intended.
3. Efficiency: At higher loads, the converter achieved a peak efficiency of 94.2%, which is close to the simulated value of 95%.
4. Thermal performance: temperature measurements for the controller and the MOSFETs confirmed that the design handled heat effectively.
5. During dynamic load regulation, the circuit exhibited unexpected overshoot behaviour that perhaps indicates inadequate feedback network compensation. Dynamic testing should be investigated further and remains an extension possibility of this project.

## 7 BIBLIOGRAPHY

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- [1] "LM3150," *Ti.com*, 2015.  
[https://www.ti.com/product/LM3150?keyMatch=LM3150&tisearch=universal\\_search&usecase=GPN-ALT](https://www.ti.com/product/LM3150?keyMatch=LM3150&tisearch=universal_search&usecase=GPN-ALT)
- [2] "Power Designer," *Ti.com*, 2024. <https://webench.ti.com/power-designer/switching-regulator/customize/4>
- [3] R. Keim, "Understanding Switch-Mode Regulation: The Buck Converter," *Allaboutcircuits.com*, Jul. 30, 2023. <https://www.allaboutcircuits.com/technical-articles/understanding-switch-mode-regulation-the-buck-converter/>
- [4] "Buck Converters," *learnabout-electronics.org*. <https://learnabout-electronics.org/PSU/psu31.php>
- [5] Wikipedia Contributors, "Buck converter," *Wikipedia*, Oct. 20, 2019.  
[https://en.wikipedia.org/wiki/Buck\\_converter](https://en.wikipedia.org/wiki/Buck_converter)

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